Atlantic cod (*Gadus morhua*)

Summary Scores

- **Sensitivity:** High - Very High
- **Exposure:** Moderate
- **Vulnerability:** Moderate - High

Changes from Previous Assessment(s)

Table 1: Changes to Vulnerability, Exposure, and Sensitivity scores relative to Previous Assessments. If a citation for the previous assessment is not provided, it comes from Hare et al. (2016).

Metric	Previous Assessment(s)	Current Assessment	Change
Sensitivity	Moderate	High - Very High	Increase
Exposure	High	Moderate	Decrease
Vulnerability	Moderate	Moderate - High	None - Increase
Directional Effect	Negative	Negative	None

Stocks

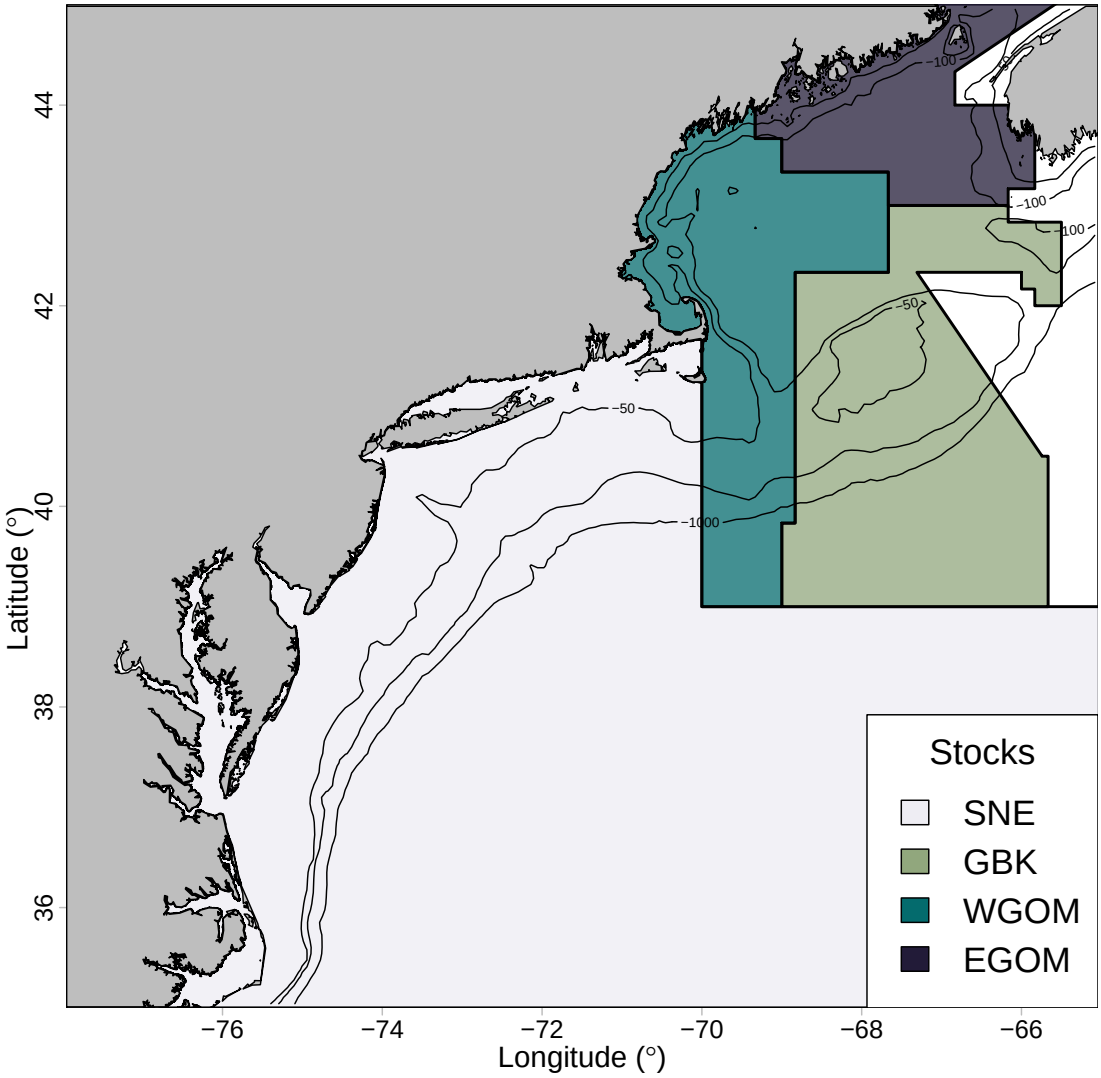


Figure 1: Atlantic cod Stocks along the Northeast Shelf

Score Descriptions:

Sensitivity:

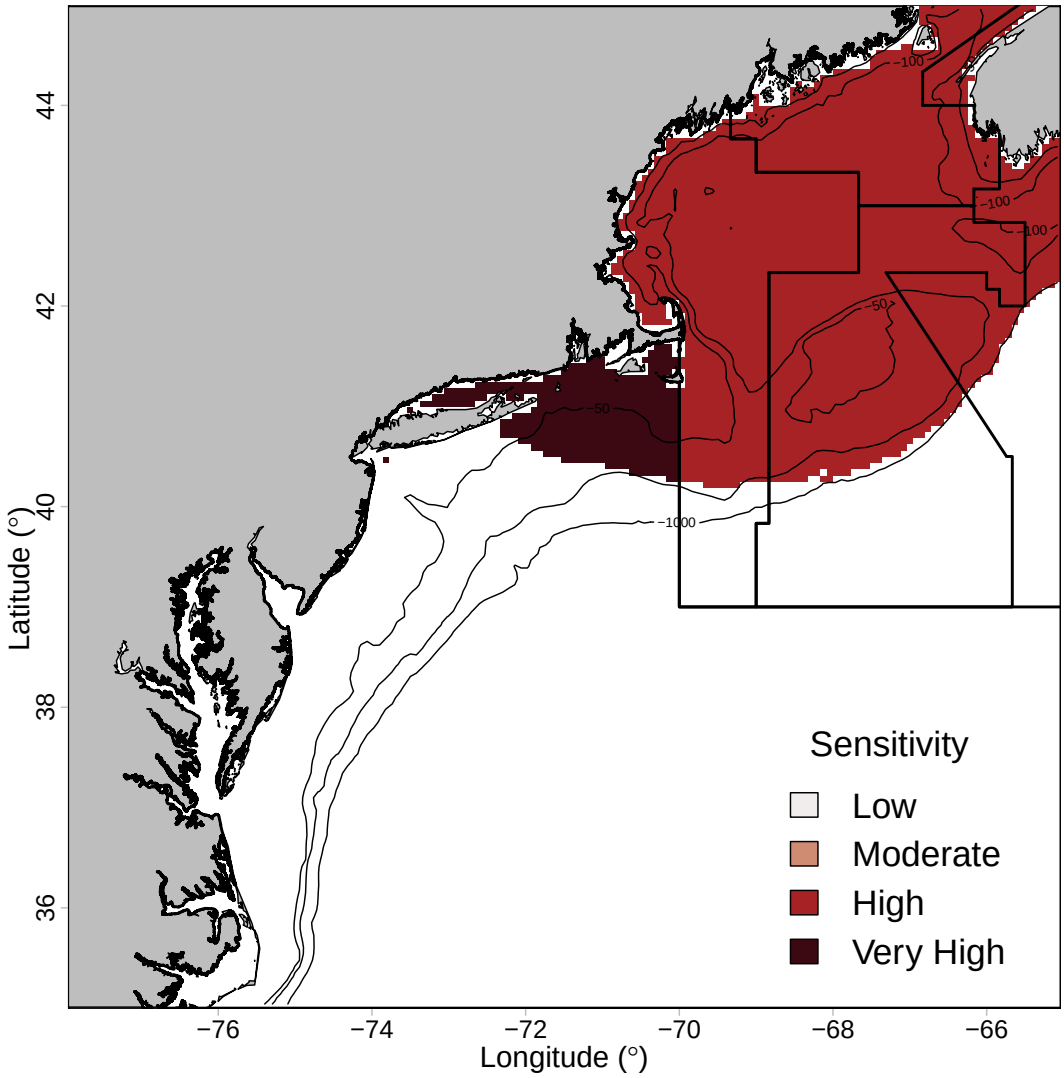


Figure 2: Atlantic cod Sensitivity on the Northeast Shelf

Of the four cod stocks, three (**Eastern Gulf of Maine, Western Gulf of Maine, and Georges Bank**) received **High** sensitivity scores. The **Southern New England** stock received a **Very High** score. Certainty was highest for the Georges Bank (92%) stock, followed by the Eastern Gulf of Maine (82%), Western Gulf of Maine (66%), and Southern New England (64%) (**?@fig-sum-tab-cod**).

Stock size, complexity in reproductive strategy, and egg and larval survival and settlement requirements drove the scores across all four stocks. Spawning cycle and historical variability in temperature also scored above 3 for all stocks (**?@fig-sum-tab-cod**).

Cod are currently overfished in the Gulf of Maine (Northeast Fisheries Science Center (NEFSC) 2021). Spawning is coupled with copepod production (Brander and Hurley 1992) and may be impacted by warming temperatures (Behan et al. 2023; Drinkwater 2005). Warming temperatures may also reduce

adult fitness, and thus reduce egg quality (Klein, Smith, and Kritzer 2016). Recruitment success is linked to retentive circulation patterns in the Gulf of Maine and on Georges Bank (Cong, Sheng, and Thompson 1996; Lough and Bolz 1989), which can be altered by large scale climate patterns and overall regional warming (Pershing et al. 2015).

Atlantic cod sensitivity has increased since the last assessment. Atlantic cod were previously scored as Moderate (Hare et al. 2016). Stock size/status and egg and larval survival and settlement requirements (formally early life history requirements) drove the scores in both assessments.

Exposure:

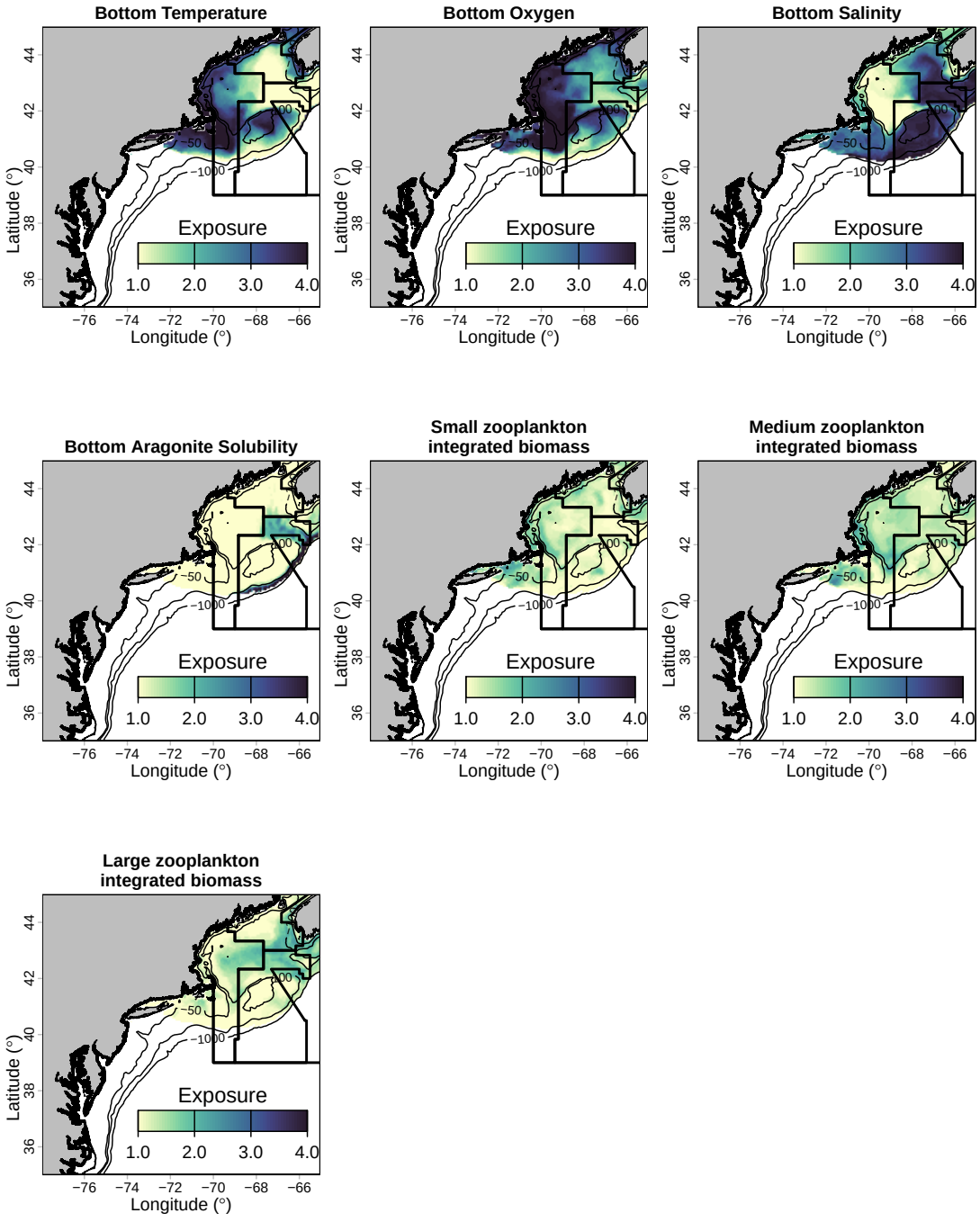


Figure 3: Maps of each Exposure Factor for Atlantic cod. Exposure was calculated between 2014-2023 and 2025-2035

Mean (+/- standard deviation) exposure scores within all four cod stocks were **Moderate** when all exposure factors were considered (**?@fig-sum-tab-cod**). Across the species range, exposure scores were

greatest along the coastline, southeast of Cape Cod, and over Georges Bank (Figure 4). These patterns were driven by high exposure to changes in bottom temperature, salinity, and oxygen. High bottom temperature exposure south of Cape Cod and Long Island drove the higher scores observed in the Western Gulf of Maine and Southern New England Stocks (Figure 3).

In the previous assessment, cod received a High exposure score, driven by high exposure to increased sea surface temperatures and ocean acidification (Hare et al. 2016).

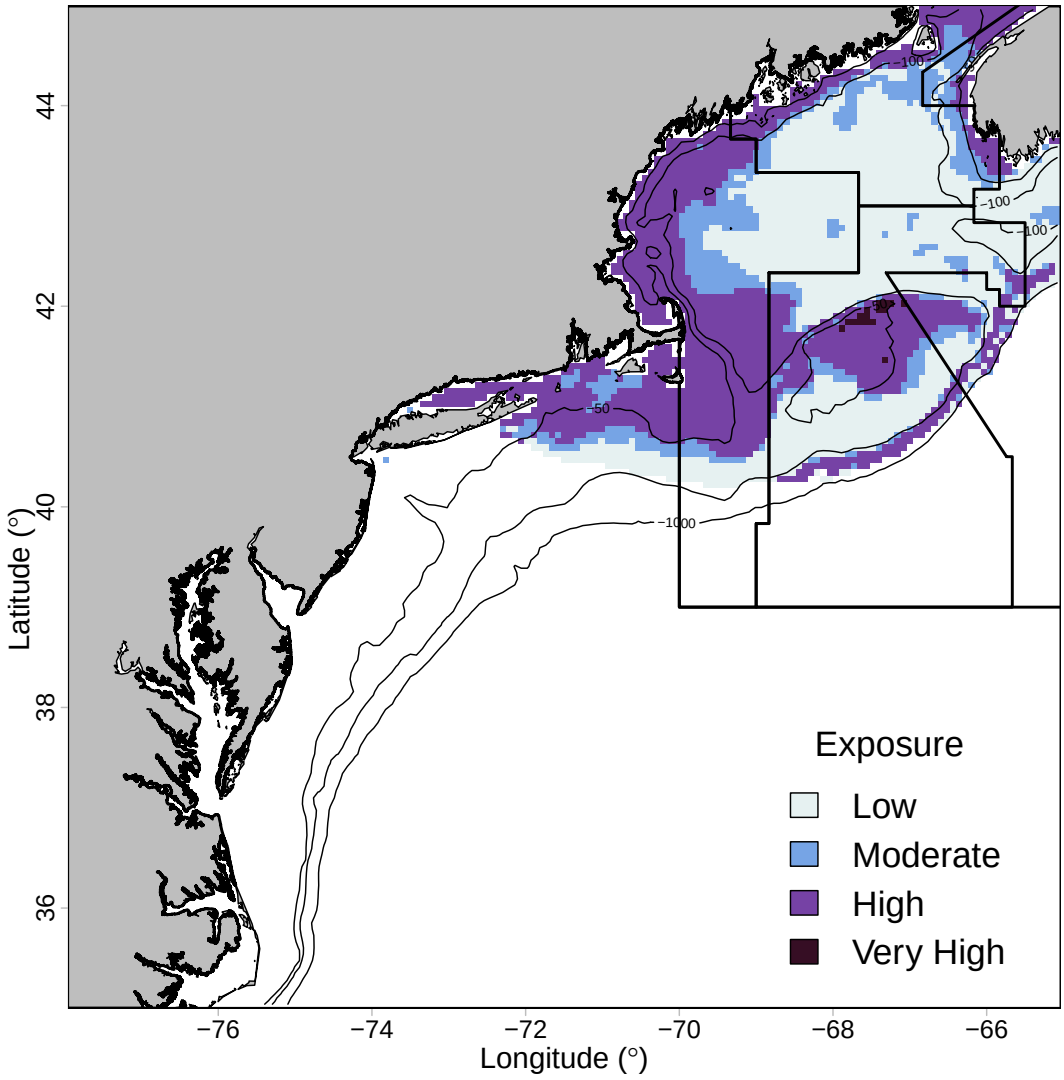


Figure 4: Total Exposure on the Northeast Shelf for Atlantic cod. Total Exposure is calculated by applying the logic rule from Hare et al. (2016) within grid cells across exposure factors (see Figure 3)

Vulnerability:

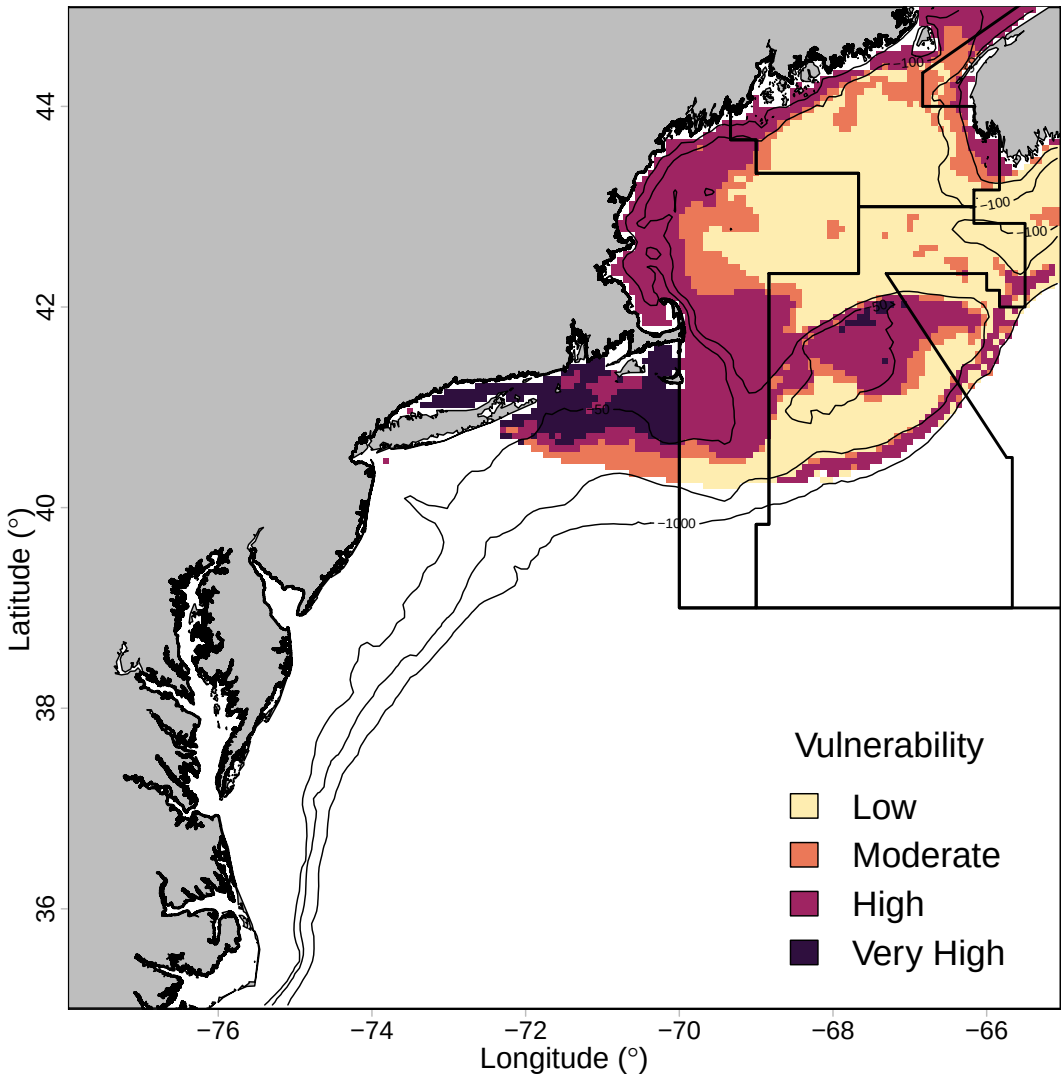


Figure 5: Vulnerability on the Northeast Shelf for Atlantic cod. Vulnerability is the product of sensitivity and exposure (see Figure 4 and Figure 2)

Mean vulnerability scores within the **Eastern Gulf of Maine, Western Gulf of Maine, and Georges Bank** stocks were **Moderate**, ranging from 5.01 to 6.75 (+/- 2.50 - 2.82). Mean scores in the **Southern New England** stock were **High** (9.8 +/- 3.09). This was driven by the higher Sensitivity score within the stock (**?@fig-sum-tab-cod**).

Very High Vulnerability scores were found south of Cape Cod, due to high exposure and high sensitivity within the Southern New England stocks. High Vulnerability scores were found along the coast of the Gulf of Maine and over Georges Bank due to high exposure, whereas exposure was Low within the Gulf of Maine basin (Figure 5).

Vulnerability scores in the previous Assessment were Moderate for Atlantic cod, similar to our results (Hare et al. 2016).

Data Quality Metrics

Life History Data Quality:

82% of the data quality scores were greater than or equal to 2 for the Georges Bank, Western Gulf of Maine, and Southern New England stocks. This value was 83% for the Eastern Gulf of Maine stock, indicating strong data quality. For all stocks, only Population Growth Rate was considered a data gap (mean attribute data quality score < 2) (**?@fig-sum-tab-cod**).

Model Confidence Score:

Experts had moderate confidence in the Atlantic cod model (mean confidence = 1.9 +/- 0.3). Experts noted that cod are often associated with rocky bottoms, which are undersampled by traditional trawl surveys used to build the models. The ensemble model AUC was 0.84, indicating good performance.

Additional Metrics

Distribution Shifts:

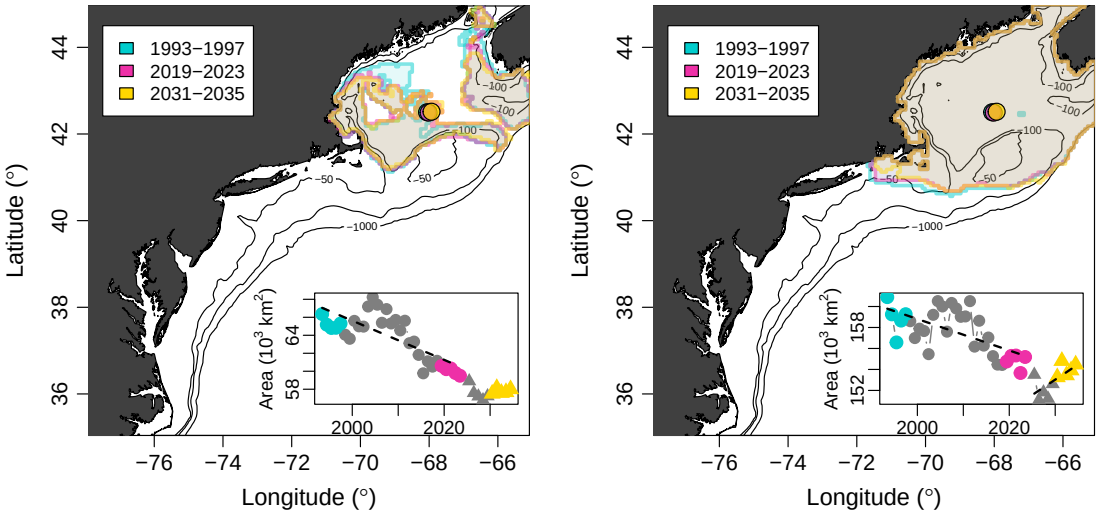


Figure 6: Distribution shifts on the Northeast Shelf for Atlantic cod. Polygons illustrate core habitat (left) and species range (right) for five-year average periods. Points illustrate weighted centroid, calculated as the average position with species distribution models as weights. Inset timeseries illustrate the shift in core and habitat range area over time. Circles represent points from the MOM6 hindcast; triangles represent points from the MOM6 forecast. Dashed lines represent significant linear regressions

Note: For an animated version of this plot, please see the online version of this document.

The weighted distribution centroid shifted 3.8 km ESE between 1993-1997 and 2019-2023. Declines in latitudinal range led to significant losses in core and total habitat area (Figure 6). This was driven by the southern edge of the species' range moving further north (Figure 6).

In the next ten years, the distribution weighted centroid is expected to shift a further 7.3 km ENE. Cod ranges are predicted to expand between 2025-2035, but will remain similar to the 2019-2023 range (Figure 6).

In the previous assessment, cod were given a High Potential for Distribution Change score (Hare et al. 2016).

Table 2: Distribution shifts based on five-year average distributions. Changes are calculated relative to the previous time period.

Time Period	Weighted Centroid Shift (km & bearing)	Core Habitat Area (change) (km2)	Habitat Total Area (change) (km2)	Latitude Range (change) (deg)	Change in Longitude Range (deg)
1993-1997	NA	65,441	159,006	3.43	4.97
2019-2023	3.8 km ESE	60,182 (-5,258)	154,872 (-4,133)	3.3 (-0.12)	4.97 (0)
2031-2035	7.3 km ENE	57,770 (-2,412)	153,955 (-917)	3.3 (0)	4.97 (0)

Directional Effect:

All four stocks of Atlantic Cod will likely be negatively impacted by future climate change (100% certainty in expert scores). This is similar to the previous Assessment (Hare et al. 2016). Continued warming will likely drive declines in recruitment and shifts in spawning phenology.

Climate Effects on Abundance and Distribution

Recent warming in the Gulf of Maine relating to a northward shift in the Gulf Stream and changes in the Atlantic Multidecadal Oscillation and Pacific Decadal Oscillation led to reduced Atlantic cod recruitment and increased mortality (Pershing et al. 2015). Rising temperatures may affect Atlantic cod spawning locations and timing (Behan et al. 2023). Drinkwater (2005) suggested Atlantic cod spawning grounds will experience a shift northward, with spawning timing occurring later in the season as a response to warming ocean temperatures. Climate change may also increase in natural mortality in adult Atlantic cod due to declines in fish condition (Behan et al. 2023). Drinkwater (2005) predicted large-scale declines in Atlantic cod in the Gulf of Maine if bottom temperatures increase above 12°C, either through changes in distribution to colder waters or natural mortality. Reduced fitness in Atlantic cod adults from climate impacts may result in lower quality eggs (Klein, Smith, and Kritzer 2016); however, there may be some capacity to adapt to slower rates of warming (Langan et al. 2020).

Life History Synopsis

Cod in the Northwest Atlantic range from Greenland to Cape Hatteras, NC, with a majority of biomass in the Gulf of Maine and on Georges Bank (Behan et al. 2023). Spawning timing varies by location - later in the year with increasing latitude - although peaks generally occur in the winter and spring (Lough 2004) in water temperatures between 5-7°C. In spring, Atlantic cod spawning matched the variation in peak abundance of *Calanus finmarchicus*. In fall, cod spawned in some, but not all, areas where *Calanus* were abundant. This is consistent with the hypothesis that cod spawning is coupled to copepod production (Brander and Hurley 1992). Atlantic cod eggs are pelagic and hatch after 8-60 days, varying with temperature (Hardy 1978). Recruitment success increases when dispersion of larvae by currents is

reduced (Cong, Sheng, and Thompson 1996). On Georges Bank, a clockwise gyre circulation pattern tends to retain and isolate eggs and larvae (Lough and Bolz 1989). Transformation from larvae to the juvenile stage occurs at sizes greater than 20 mm, where they then move to the bottom (Lough 2004). Early juveniles are dependent on gravel substrates or grass beds in nearshore nurseries to favor survival through predator avoidance and increased food availability (Lough and Bolz 1989). In the Gulf of Maine, juveniles prey on mollusks and other invertebrates, while adults consume a greater proportion of fish including herring, redfish, mackerel, and other cod (Bowman 1975). Adult cod have few predators, other than large sharks and seals (Neuenhoff et al. 2019). Recent modeling studies illustrate growing populations of grey seals, harp seals, and dogfish are a major barrier to cod population recovery (Behan et al. 2023). Adults in the Newfoundland stock migrate, leaving overwintering areas in deep oceanic waters and follow tongues of deep, relatively warm, oceanic waters ("highways") across the shelf to summer feeding areas nearshore. They then move northward along the Newfoundland coast in late summer, and eventually return to wintering areas (Lough 2004). Previous research suggested that in the middle part of their range, Atlantic cod only undertake minor seasonal movements in reaction to changing temperatures. However, more recent research suggests that there are inshore/offshore migrations that correspond to spawning and feeding activity in the Gulf of Maine (Zemeckis et al. 2017). As of 2025, Atlantic cod are divided into 4 stock units (Western & Eastern Gulf Maine, Georges Bank, and Southern New England) based on spawning locations, larval transport pathways, and genetic markers (New England Fishery Management Council (NEFMC) 2025). All stocks are managed under the New England Fishery Management Council's Northeast Multispecies Fishery Management Plan (Klein-MacPhee 2012). The stock is overfished and overfishing is occurring in the Gulf of Maine (Northeast Fisheries Science Center (NEFSC) 2021).

Supplemental Scores

Exposure Timeseries

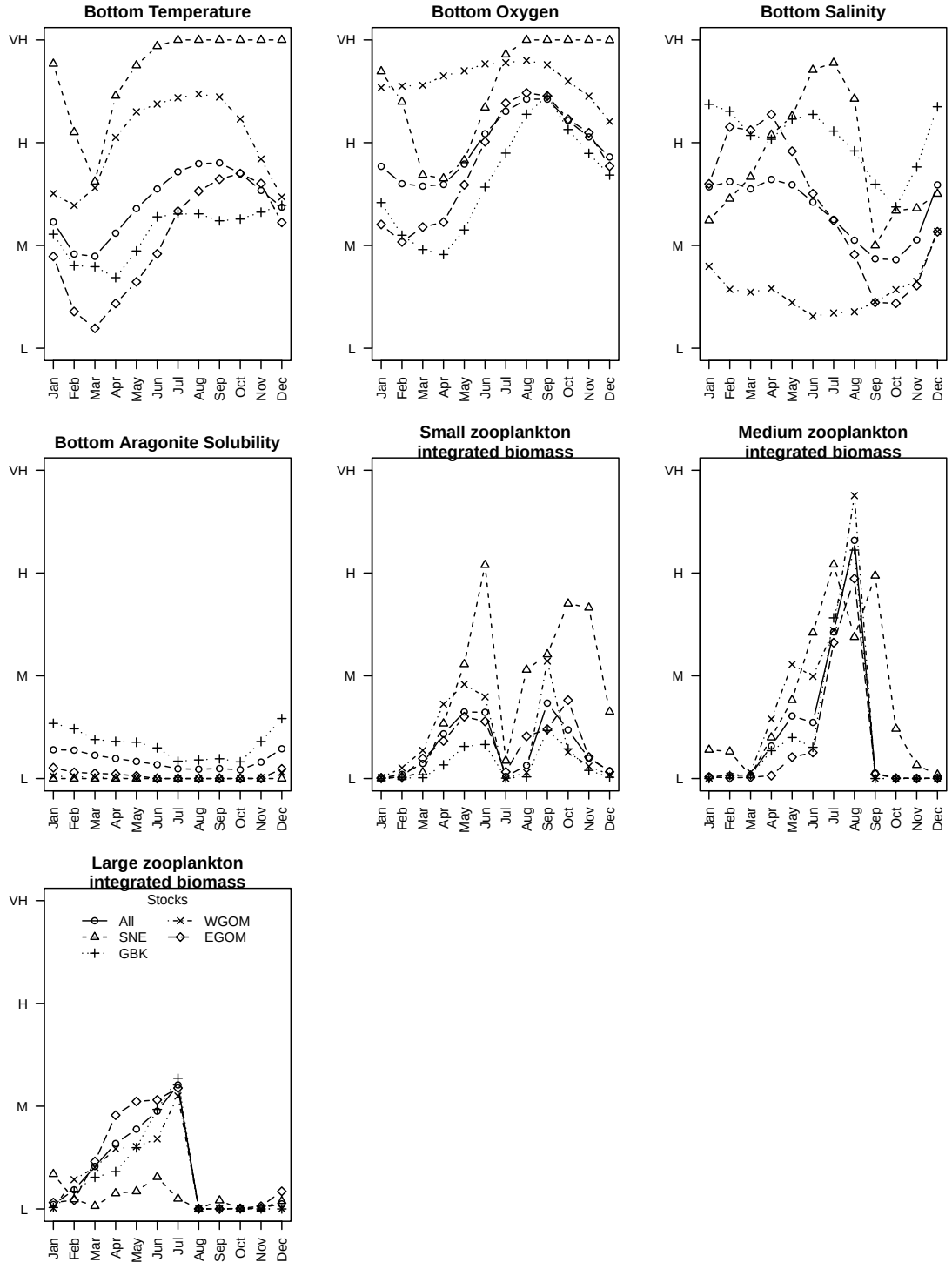


Figure 7: Monthly Timeseries of each Exposure Factor for Atlantic cod. Exposure was calculated between 2014-2023 and 2025-2035

While total exposure was relatively consistent across months for cod across their entire range, there was variability throughout the year within the different stocks (Figure 8). Georges Bank, Western Gulf of Maine, and Southern New England stocks had the highest variability. Exposure within the Georges Bank and Western Gulf of Maine stocks ranged from Low to High, whereas exposure ranged from Moderate to Very High in Southern New England. Exposure was generally highest in the summer months, due to high bottom temperature, oxygen and zooplankton biomass exposures (Figure 7).

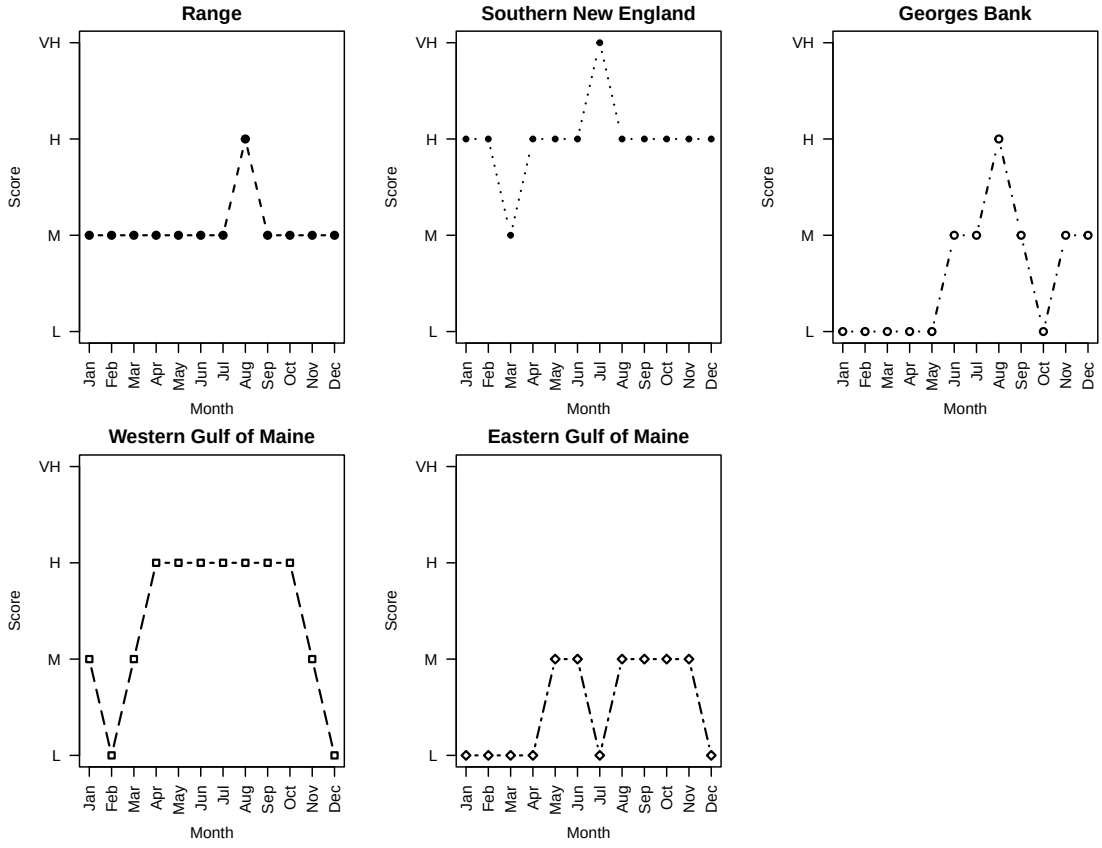


Figure 8: Timeseries of Total Exposure for Atlantic cod

Exposure due to Significant Factors

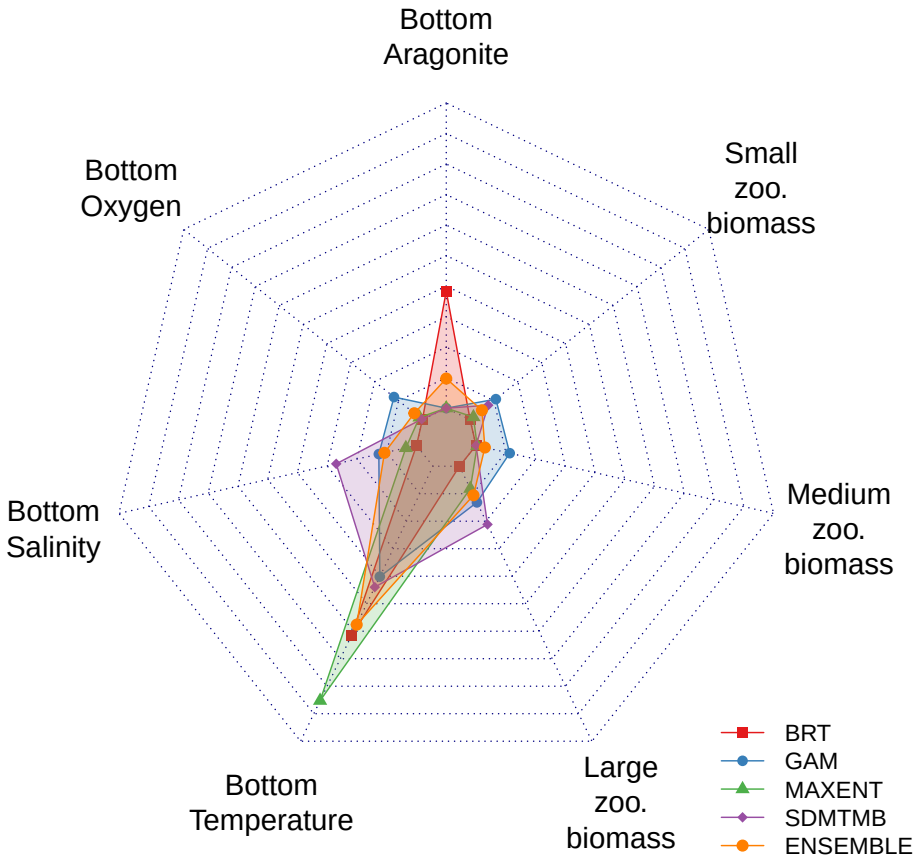


Figure 9: Dynamic Factors driving Atlantic cod distributions. The closer the polygon is to the edge of the figure, the more important the variable is. Each ring represents 10% variance explained. Factors must explain at least 10% of variance in the ensemble model to be used to calculate total exposure.

Bottom temperature, bottom salinity, and large zooplankton biomass were the most important dynamic variables in the Atlantic Cod species distribution model (Figure 9). Cod are piscivores, feeding on other small fishes that likely prey on zooplankton (Bowman 1975). Spawning is closely linked to temperature (Lough 2004). When only important factors were used to calculate exposure, exposure in the Eastern and Western Gulf of Maine stocks, as well as the Georges Bank stock, decreased to Low (Figure 10; **?@fig-sum-imp-cod**). The mean exposure score only remained Moderate in the Southern New England stock (Figure 10; **?@fig-sum-imp-cod**).

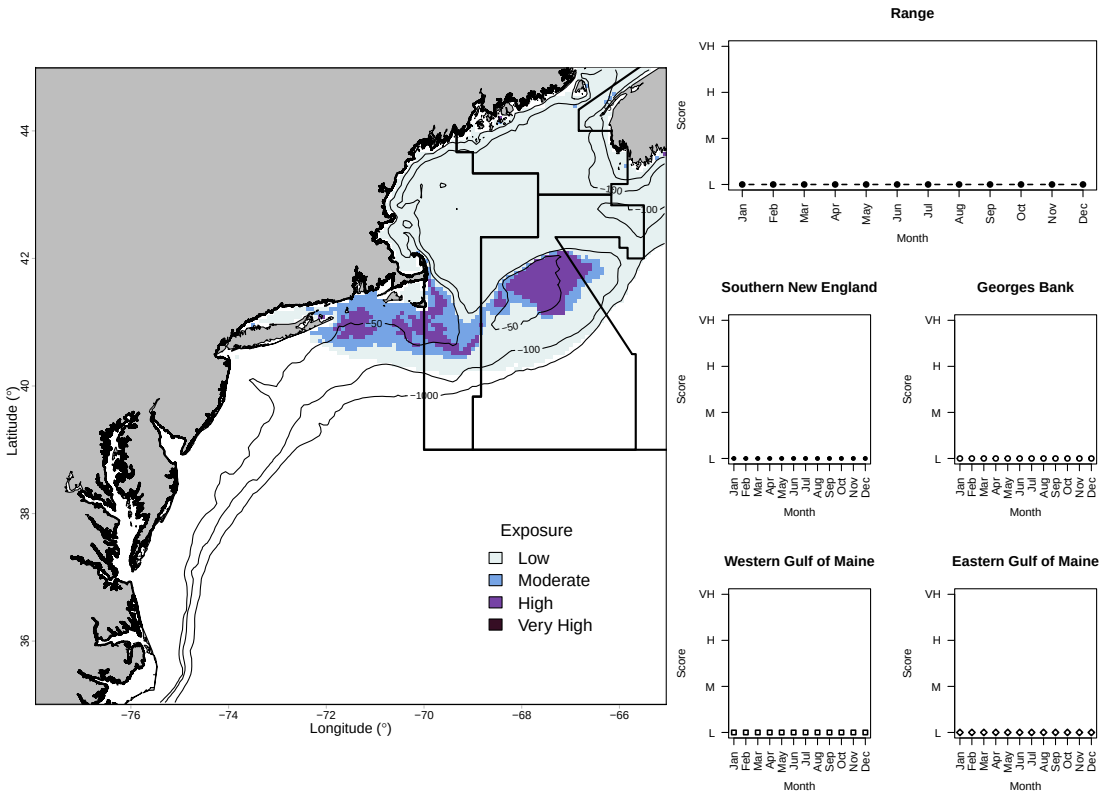


Figure 10: Total Exposure Map and Timeseries when Exposure was calculated with only Exposure Factors critical to Atlantic cod distributions

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